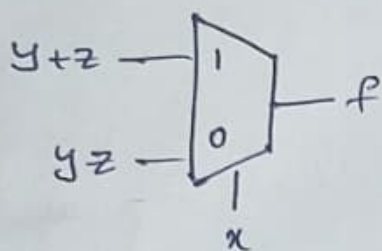


① Whenever majority of x, y, z are 1; $F(x, y, z) = 1$

$$\Rightarrow F(x, y, z) = xy + yz + zx + xyz = xy + xyz + \bar{x}yz + zx + xyz$$



$$= x(y+z+yz) + \bar{x}yz$$

$$= x(y(1+z) + z) + \bar{x}yz$$

$$= x(y+z) + \bar{x}yz$$

$$\Rightarrow \boxed{P = \text{OR} \quad Q = \text{AND}}$$

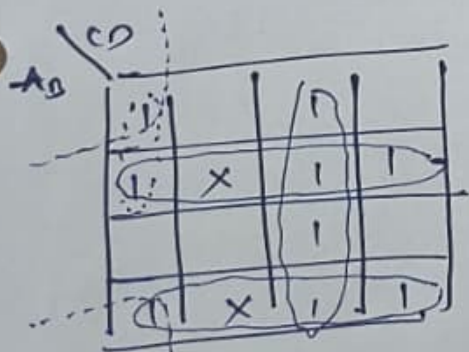
② a) $f_2 = f_1 + f_3 = \Sigma m(1, 6, 8, 15)$

$$f_1 \cdot f_2 = \Sigma m(1, 6, 8, 15) - \Sigma m(1, 4, 15)$$

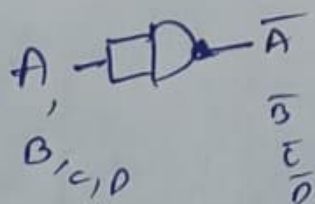
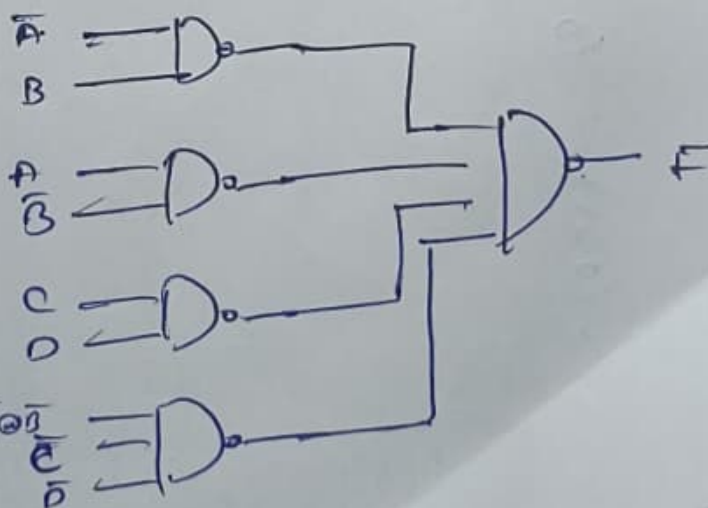
$$= \Sigma m(8) \quad (\because 4, 5, 6, 7 \text{ should definitely be '0' as they are in } f_1)$$

$$\Rightarrow \boxed{f_2 = \Sigma m(8) + d(0, 1, 2, 3, 9, 10, 11, 12, 13, 14, 15)}$$

b) $\Pi(1, 2, 12, 13, 14) = \Sigma(0, 3, 4, 6, 7, 8, 10, 11, 15)$
 $+ d(5, 9)$



$$F = \overline{A}B + A\overline{B} + CD + (\overline{A}\overline{C}\overline{D} \text{ (or) } \overline{B}\overline{C}\overline{D})$$



CD

$$K_1 \overline{A}B + A\overline{B} + CD + (\overline{A}\overline{C}\overline{D} \oplus \overline{C}\overline{D})$$

4D
= 10

(3)

	A	B	C	D	F
1	0	0	0	1	0
2	0	0	1	0	1
4	0	0	0	0	0
7	0	1	1	1	0
8	1	0	0	0	0
11	1	0	1	1	1
13	1	1	0	1	1
14	1	1	1	0	0

$$F: \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}D + A\overline{A}C\overline{D} + AB\overline{C}D = SSOP.$$

(4)

Positive number in 2's complement $\Rightarrow$ MSB = 0

$$\Rightarrow X_3 > 0$$

$$\Rightarrow X_2 \times 1 \times 0 \times 0 \times 1$$

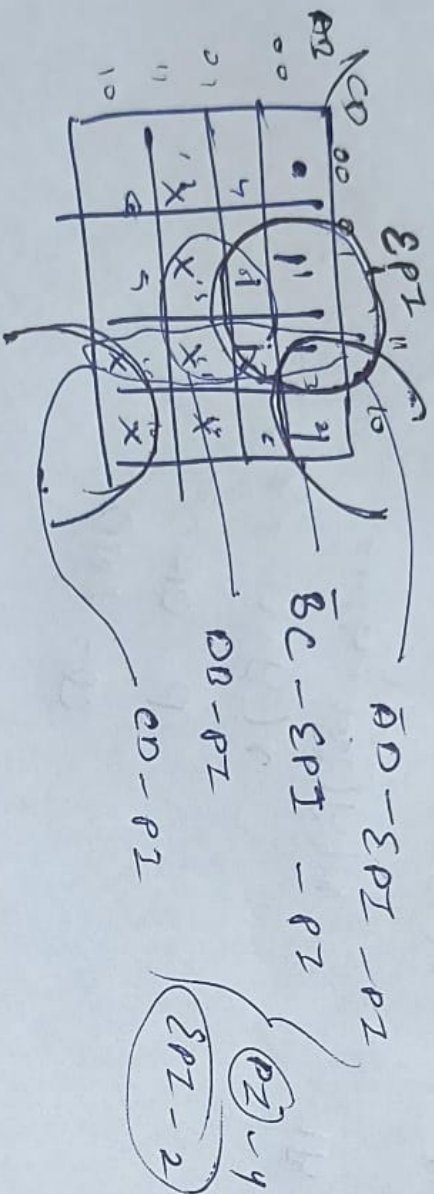
$$2^4 \times 2^3 \times 2^2 + 1$$

∴ remainder = 1

When divided by $(100)_2$

$$= 4$$

$$\Sigma m(1,2,3,5,7) + d(10,11,12,13,14,15)$$



EP1: At least one 1's grouped only once.

Not used 1's PI
provided as group getting 1 by x

Yes used 1's EP1

PI - Product term obtained by grouping 1's (usual care) if helpful that can't be expanded further.

Q

$$A \odot \bar{B} = A \oplus B = x$$

$$x \odot \bar{x} \odot y \odot \bar{y} = (x \odot \bar{x}) \odot (y \odot \bar{y})$$

$$A \odot B = \overline{A \oplus B}$$

$$= 0 \odot 0$$

$$= 1$$

$$x \odot \bar{x} = x \odot 0 = \overline{x \oplus 0} = \bar{x}$$

$$x \odot 0 = x \oplus 0 = x$$

$$(0.65)_{10} = 0.65 \times 8 = \underline{5.2}$$

$$(0.2)_{10} = 0.2 \times 8 = \underline{1.6}$$

$$= 0.6 \times 8 = \underline{4.8}$$

$$= 0.8 \times 8 = \underline{6.4}$$

$$\therefore (0.65)_{10} = (0.5146\ldots)_8$$

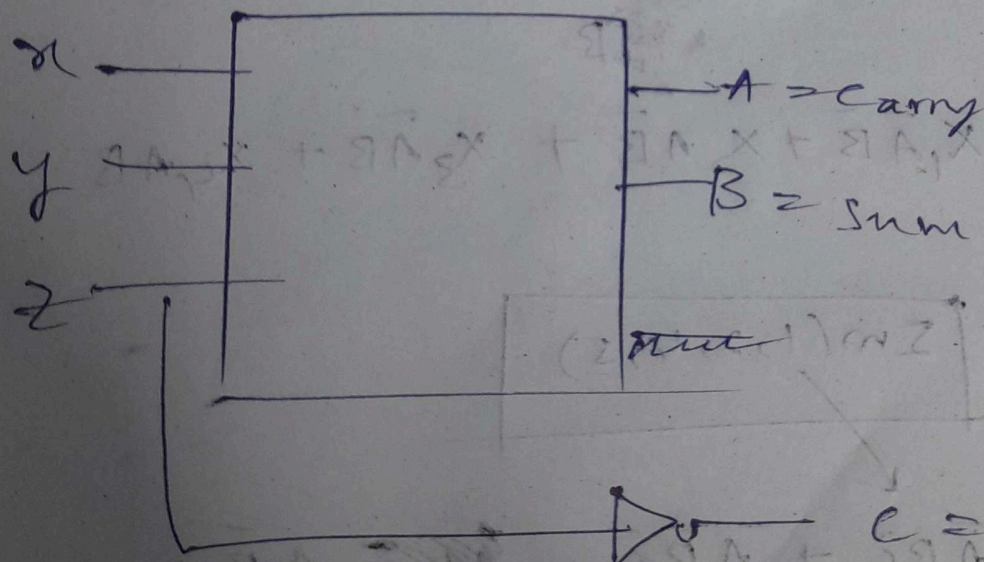
$$\begin{array}{cccccccc} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & \rightarrow \text{Gray} \\ \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \\ 1 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & \rightarrow \text{binary} \\ 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$$= (492)_{10} = (0100 \ 1001 \ 0010)_8$$

$\bar{B}$								
B								
$\# S_i$	$\bar{A}\bar{B}\bar{D}$	$\bar{A}\bar{C}\bar{D}$	$\bar{A}\bar{C}\bar{D}$	$\bar{A}\bar{C}\bar{D}$	$\bar{A}\bar{C}\bar{D}$	$\bar{A}\bar{C}\bar{D}$	$\bar{A}\bar{C}\bar{D}$	$\bar{A}\bar{C}\bar{D}$
	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{B}$	(0)	(1)	(2)	3	8	(9)	10	11
B	4	(5)	6	(7)	(12)	13	(14)	(15)

$\bar{B}, 1, \bar{B}, B, B, \bar{B}, 0, B$

(10)
(a)



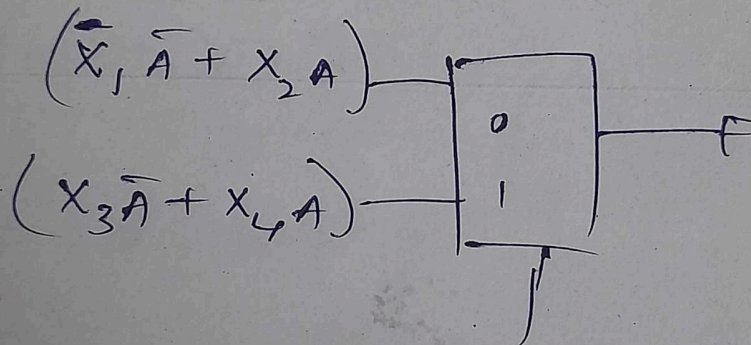
b) Draw truth table

=>

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{w}\bar{x}$	1	1	x	x
$\bar{w}x$	0	0	1	1
$w\bar{x}$	0	1	1	1
wx	1	1	1	0

Ans: $\bar{x}\bar{y} + xy + wz = \boxed{Ans = 6}$

11.5)



=> $F = X_1\bar{A}\bar{B} + X_2A\bar{B} + X_3\bar{A}B + X_4AB$

we have $\boxed{\Sigma m(1, 2, 4, 5)}$

$\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + \cancel{A\bar{B}C} + A\bar{B}C$

Compare => $X_1 = C$; $X_2 = \bar{C}$; $X_3 = \bar{C} + C$

$\boxed{X_4 = 0} = 1$